



Book I. Prop. XXVII. Theor. *If a straight line (—) meeting two other straight lines (— and —) makes with them the alternate angles (▲ and ▼; ◐ and ◑) equal, these two straight lines are parallel. If*

— be not parallel to — they shall meet when produced.

If it be possible, let those lines be not parallel, but meet when produced; then the external angle ▼ is greater than ▲ (pr. I.16), but they are also equal (hyp.), which is absurd: in the same manner it may be shown that they cannot meet on the other side; ∴ they are parallel.

Q. E. D.